

COMPUTE ALLOCATION REPORT

AWS Cloud Credit for Research

Application-based AWS credits for academic research — scales with justification

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Program Overview

Amazon Web Services has run an academic credit program under several names since 2013. The original *AWS Cloud Credit for Research* accepted open proposals against a public application form, but in 2022 AWS retired the public intake and folded the program into a smaller set of more tightly managed channels. As of the 2024–2026 cycle, the practical paths into AWS research credits are three. The first is the *AWS Promotional Credits for Research* track, which is functionally the successor to the old Cloud Credit for Research program but is now routed through institutional AWS contacts and account managers rather than a public form. The second is the *AWS Research and Educational Program* request, a managed-account-level credit grant for accredited institutions that an AWS account manager (or a sponsoring AWS Solutions Architect) submits internally on behalf of the researcher. The third is the *AWS IMAGINE Grant* for nonprofits, which issues larger awards (frequently \$50k–\$150k in credits plus professional services) to mission-driven organizations including university research centers; the IMAGINE Grant runs an annual public RFP and is the most discoverable of the three.

Two adjacent programs are worth knowing because they are frequently conflated with the credit programs but are operationally distinct. The *AWS Generative AI Innovation Center* runs academic engagements that pair a research group with an AWS applied scientist team for a multi-week prototyping sprint; credits may be bundled, but the deliverable is consulting time rather than a pure compute allocation, and intake is by referral. The *AWS Diagnostic Development Initiative*, launched in 2020 with an initial \$20M commitment for COVID-era diagnostics research, was extended through public-health and infectious-disease use cases and remains active as of early 2026 with a narrower scope; it issues credits and technical support against disease-surveillance and diagnostics proposals. *AWS Build On* is a separate educational tier aimed at student builders and hackathons, typically issuing \$25–\$300 in credits per participant rather than research-grade allocations.

Hardware Specs

The table below lists the AI-relevant accelerated instance families on AWS as of the 2026 EC2 catalog. Per-GPU on-demand and 1-year standard-reserved prices are computed by dividing the published instance list price (us-east-1) by the GPU count, rounded to the nearest dollar. The TFLOPS column is dense FP16/BF16 tensor throughput without sparsity, directly comparable to the 989 TFLOPS H200 baseline used elsewhere in this report. Trainium and Inferentia entries report BF16 dense throughput because those parts have no FP16 tensor mode in the conventional sense.

Instance	GPUs	VRAM (GB)	FP16/ BF16 TFLOPS	BW (TB/s)	On-de- mand \$/ GPU-hr	1-yr RI \$/ GPU-hr
p5.48xlarge (H100)	8	80	989	3.35	\$12.29	\$7.57
p5e.48xlarge (H200)	8	141	989	4.80	\$11.06	\$5.07
p5en.48xlarge (H200)	8	141	989	4.80	\$11.57	\$5.71
p4d.24xlarge (A100 40)	8	40	312	1.55	\$4.10	\$2.55
p4de.24xlarge (A100 80)	8	80	312	2.00	\$5.12	\$3.20
g6e.48xlarge (L40S)	8	48	362	0.86	\$3.04	\$1.92
g6.48xlarge (L4)	8	24	121	0.30	\$1.78	\$1.10
g5.48xlarge (A10G)	8	24	125	0.60	\$2.06	\$1.30
trn1.32xlarge (Trainium1)	16	32	190	0.82	\$1.36	\$0.85
trn2.48xlarge (Trainium2)	16	96	667	2.90	\$3.12	\$1.95
inf2.48xlarge (Inferentia2)	12	32	191	0.82	\$1.05	\$0.66

The dominant cost-per-FLOP winners for training are Trainium2 and the H200-based p5e family, which is why most credit-funded H100/H200 training campaigns in 2025–2026 target p5e specifically rather than the older p5 (H100) or p4d (A100) families. The g6e family with L40S is the most cost-effective option for fine-tuning and inference workloads that fit in 48 GB per GPU.

Credit Amount and H200-Hour Equivalence

Research-tier awards in 2024–2026 cluster around \$5k–\$20k for first-time PI requests, \$25k–\$50k for follow-on awards backed by publications, and \$50k–\$100k+ for IMAGINE grants or institutionally-sponsored multi-year engagements. The *H200-hour equivalence* of a credit award is highly sensitive to which purchasing mode the researcher uses. The table below converts each award tier to H200-hours on p5e (the dominant H200 instance) and on p4d (A100 reference) under on-demand, 1-year reserved, and spot pricing. Spot prices are taken at the 2025–2026 average of roughly 60% off on-demand for p5e (when capacity is available) and roughly 70% off for p4d.

Award	p5e on-dem.	p5e 1-yr RI	p5e spot	p4d on-dem.	p4d spot
\$5,000	452 hr	986 hr	1,131 hr	1,220 hr	4,065 hr
\$10,000	904 hr	1,972 hr	2,262 hr	2,439 hr	8,130 hr
\$25,000	2,261 hr	4,931 hr	5,654 hr	6,098 hr	20,325 hr
\$50,000	4,521 hr	9,862 hr	11,308 hr	12,195 hr	40,650 hr
\$100,000	9,042 hr	19,724 hr	22,617 hr	24,390 hr	81,301 hr

Worked Conversions

A \$5,000 award spent at the p5e on-demand list price of \$11.06/GPU-hr yields $\frac{5000}{11.06} \approx 452$ H200-hours. The same \$5,000 spent against a 1-year standard-reserved p5e at \$5.07/GPU-hr (the lowest no-upfront RI rate published in 2026) yields about 986 H200-hours, more than doubling the throughput per dollar at the cost of a 12-month commitment that most credit awards cannot fully fund. Spot capacity on p5e, when available, sits near \$4.42/GPU-hr (a roughly 60% discount on on-demand) and yields about 1,131 H200-hours per \$5,000, but availability is thin in the popular us-east-1 and us-west-2 regions and preemption mid-job is common. A \$25,000 award on p5e on-demand is roughly $\frac{25000}{11.06} \approx 2261$ H200-hours, which is enough to fine-tune several 70B-class models with LoRA or to do a single 7B-parameter pretraining run on a few tens of billions of tokens at realistic utilization. The same \$25,000 stretched onto p4d spot yields roughly 20,325 A100-hours of compute, equivalent to about 6,400 H200-hours after the FP16-throughput correction ($20325 \times \frac{312}{989} \approx 6411$), which is the largest effective allocation any of the three tiers produces and the standard recipe for running large fine-tuning sweeps on academic credits.

Approval Timeline

The current AWS process is slower and less predictable than the NSF-side programs. A typical Promotional Credits for Research request moves through three steps. First, the PI contacts an institutional AWS account manager (most R1 universities have one) or a regional AWS Public Sector representative. Second, the PI submits a 2–3 page proposal that includes the research question, an architecture sketch, a line-itemed AWS budget keyed to specific SKUs, the AWS account ID that will receive the credits, and a one-page PI CV. Third, the account manager routes the request internally to AWS Research Programs for review. End-to-end turnaround is typically 4–8 weeks for a \$5k–\$25k request and 8–12 weeks for a \$50k+ request, with longer queues around AWS re:Invent (late November through January) and the end of AWS’s fiscal year (June–July). IMAGINE Grant decisions follow the public RFP cycle and are announced annually, with awards typically issued 6–10 weeks after the application deadline.

Eligibility

Eligible applicants are faculty, postdocs, and graduate students at accredited academic institutions and qualifying nonprofit research organizations worldwide, though U.S.-based institutions move through the process noticeably faster. There is an explicit open-publication expectation. Resulting work should be submitted for peer review or posted to a public preprint server, and AWS asks for an acknowledgment in the resulting paper. Credits are deposited as time-limited promotional credits on the requester’s AWS account and almost always expire 12 months from the issue date with no rollover; unused credits simply

lapse. Credits cannot be transferred between accounts, cannot be applied retroactively to past usage, and cannot be redeemed for cash or support contracts.

Things to Know

Several practical points materially affect how to use AWS credits. First, *credits do not cover every SKU*. AWS Marketplace third-party software (including some packaged ML container images), AWS Support plans above Basic, Reserved Instance upfront payments, and certain premium services (notably some Outposts and dedicated-host SKUs) are excluded; the exclusion list is published in the credit terms and should be cross-checked before budget commitment. Second, *egress fees are the silent killer*. Pulling a multi-terabyte trained model or dataset out of S3 to an on-prem cluster at the published \$0.09/GB egress rate can consume 20–40% of a small credit award; researchers who plan to keep artifacts inside AWS (S3 to SageMaker to EC2) avoid this almost entirely.

Third, *quota for p5, p5e, p5en, and p4d is gated*. New AWS accounts have a default vCPU quota of 0 for the relevant capacity-block families, and a Service Quotas request must be filed and approved before the first instance can launch. Approval typically takes 1–3 weeks, occasionally longer when global H100/H200 supply is tight, and the request must justify the requested concurrent vCPU count against the project plan. Fourth, *spot p5e availability is genuinely thin*. On many days in 2025–2026, p5e spot was simply not available in us-east-1 or us-west-2 regardless of bid; the workable fallbacks are p5 (H100) spot, p4d (A100) spot, or scheduled *Capacity Blocks for ML*, which are reservation-style purchases priced between on-demand and RI and typically available 1–14 days out. Finally, the *Generative AI Innovation Center* academic engagement is more consulting than pure credits. The AWS team co-develops the prototype and any bundled credits cover the prototype’s compute footprint, not an open-ended research budget; treat it as a partnership channel rather than a credit program.

Strategic Fit

For a graduate AI researcher who needs Hopper-class compute and already has a NAIRR Pilot allocation, AWS credits are the natural backup channel. More dollars tend to convert to more H200-hours than NAIRR’s bounded DGX Cloud allocation does, especially when spent on 1-year reserved p5e or on p4d spot, and the SKU catalog is broader than any single NAIRR provider. The trade-offs are the slower 4–8 week approval, the cold-start tax of the Service Quotas request, and the 12-month credit-expiration window. The recommended pattern is to apply for NAIRR Start-Up first (turnaround under three weeks), use NAIRR Research for the bulk of H100/H200 training, and submit an AWS Promotional Credits for Research request in parallel as a quota-isolated backup that becomes available for elasticity bursts and post-NAIRR continuation work. Pair the AWS account with *CloudBank* billing dashboards (or AWS-native Cost Explorer plus Budgets) to monitor burn rate against the 12-month expiration clock; a credit unspent at month 12 is simply forfeited.

Summary

Credit tier	On-demand H200-hr	1-yr RI H200-hr	Spot H200-hr	Approval	Effort
\$5k (first-time)	452	986	1,131	4–6 wk	Low
\$10k (follow-on)	904	1,972	2,262	4–6 wk	Low
\$25k (with publication)	2,261	4,931	5,654	6–8 wk	Medium
\$50k (institutional)	4,521	9,862	11,308	8–10 wk	Medium
\$100k (IMAGINE / GenAI IC)	9,042	19,724	22,617	10–14 wk	High

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

- AWS Cloud Credit for Research and successor programs: aws.amazon.com/government-education/research-and-technical-computing/cloud-credit-for-research/
- AWS IMAGINE Grant for nonprofits: aws.amazon.com/government-education/nonprofits/aws-imagine-grant-program/
- AWS Generative AI Innovation Center: aws.amazon.com/ai/generative-ai/innovation-center/
- AWS Diagnostic Development Initiative: aws.amazon.com/government-education/nonprofits/disaster-response/diagnostic-development-initiative/
- EC2 P5/P5e/P5en (H100/H200) instance and pricing pages: [/ec2/instance-types/p5](#) · [/ec2/pricing/on-demand](#)
- EC2 P4d/P4de (A100), G5/G6/G6e (A10G/L4/L40S), Trn1/Trn2 (Trainium), Inf2 (Inferentia2) instance pages: [/p4](#) · [/g6e](#) · [/trn2](#) · [/inf2](#)
- EC2 Capacity Blocks for ML: aws.amazon.com/ec2/capacityblocks/
- AWS Service Quotas (P5/P5e/P4d vCPU quota requests): docs.aws.amazon.com/servicequotas/
- Hardware specs cross-checked against NVIDIA H100, H200, A100, A10G, L4, L40S datasheets and AWS Trainium2 / Inferentia2 product briefs. On-demand pricing verified against the AWS pricing pages in us-east-1 (2024–2026); spot multipliers derived from EC2 Spot Instance Advisor historical averages.